

Computer vision

Morphological image processing

Scope

Neighbours of pixel, (neighbourhood's pixel)

Connectivity

Distances

Morphological processing

Kapitola 4 Morfologické operácie od str. 52



Basic terms

Neighborhood of Pixel (*Sousednost*)

a) 4-neighbors

b) diagonal neighbors

c) 8-neighbors

	x	
x	$p(x, y)$	x
	x	

(a)

x		x
	$p(x, y)$	
x		x

(b)

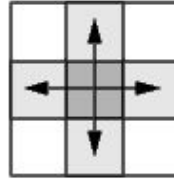
x	x	x
x	$p(x, y)$	x
x	x	x

(c)

Connectivity (*Súvislost'*)

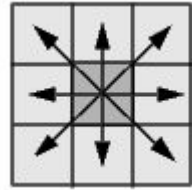
4-connected :

Two pixels p and q are
4-connected if they are
4-neighbors and $p \in V$ and $q \in V$



8-connected

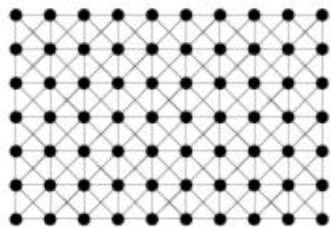
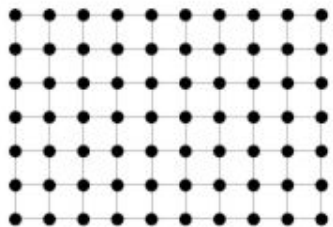
Two pixels p and q are
8-connected if they are
8-neighbor and $p \in V$ and $q \in V$



4-connectivity, 8-connectivity (4-súvislost', 8-súvislost')

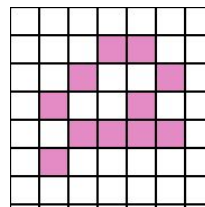
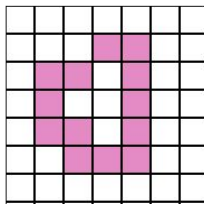
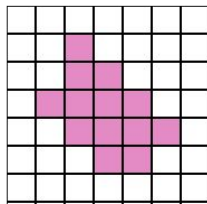
4-connected

8-connected



Examples of 4-connected pattern :

Example of 8-connected patterns :



Distance function

A metric on the points p, q is a function called the distance function.

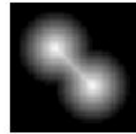
The following conditions are satisfied:

1. $D(p, q) \geq 0$ non-negativity
2. $D(p, q) = 0 \Leftrightarrow p = q$ identity
3. $D(p, q) = D(q, p)$ symmetry
4. $D(p, z) \leq D(p, q) + D(q, z)$ triangle inequality

Distance functions

Distance between pixels p and q can be defined by one of the following:

Euclidean distance: $D_E(p, q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$



City-block distance: $D_4(p, q) = |x_1 - x_2| + |y_1 - y_2|$

(The distance between two points in an image is given by the number of steps (of unit size) along a rectangular discrete grid)

0	0	0
0	1	0
0	0	0

Image

1.41	1.0	1.41
1.0	0.0	1.0
1.41	1.0	1.41

Distance transform

0	0	0
0	1	0
0	0	0

Image

2	1	2
1	0	1
2	1	2

Distance transform

0	0	0
0	1	0
0	0	0

Image

1	1	1
1	1	1
1	1	1

Distance transform

Chess-board distance: $D_8(p, q) = \max(|x_1 - x_2|, |y_1 - y_2|)$

Morphological Image Distance Transform

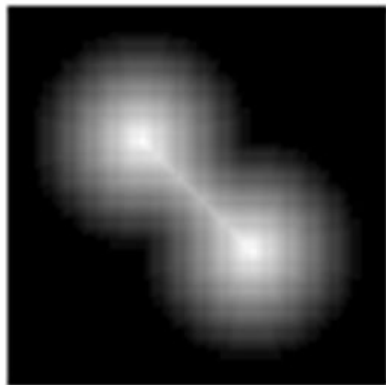
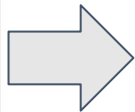
The Morphological Image Distance Transform is an image processing operation that converts a binary image into a distance map, where each pixel's value represents the minimum distance from that pixel to the nearest object boundary (or foreground/background, depending on the definition).

Distance transform - example

Applied distance transform using Euclidean distance:



Input image



Distance image

Grayscale image , binary image

grayscale image example:



binary image example:

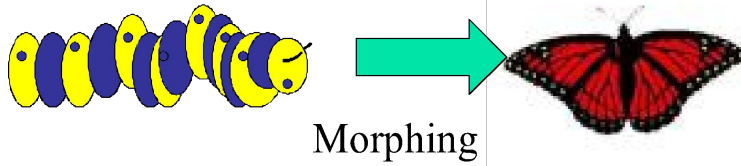


Morphological image processing

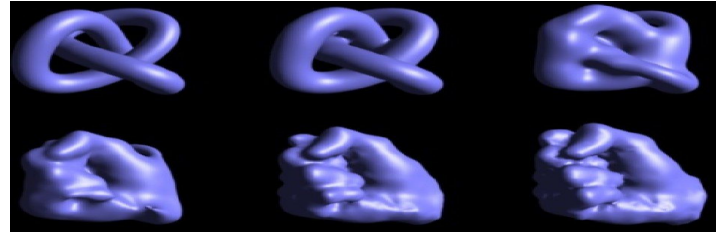
What are Morphological Operations?

Morphological operations are intended to affect the **shape** of the object.

“Morphological operations” come from the word “morphing” in biology which means changing a shape.



“Morphing” refers also to an animation technique in which one image is gradually turned into another



Morphological image processing

Morphological image processing uses information about the **neighbours in the topological surroundings** of the processing pixel.

Neighbourhood operator : **structuring element**

Used inputs:

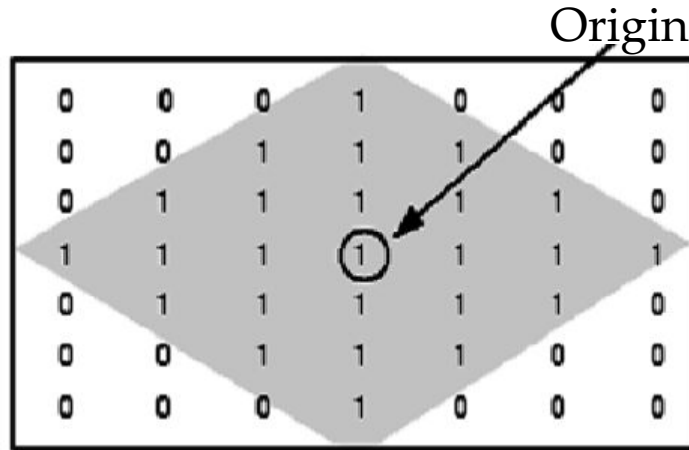
binary images, grayscale images (colour images)

The foundation of morphological processing is in the mathematically rigorous field of *set theory*, however, this level of sophistication is seldom needed.

Flat Structuring Element

A structuring element is a matrix consisting of only 0's and 1's that can have any arbitrary shape and size.

The pixels with values of 1 define the neighborhood .



Basic morphological operations: Dilation & Erosion

Dilation adds pixels to the boundaries of an (white) objects in an image, while erosion removes pixels on the object boundaries.

The number of pixels added or removed from the objects in an image depends on the size and shape of the structuring element used to process the image.

Morphological operations are typically applied to process binary images, but they are also utilized on gray-scale images.

Rules for Dilation and Erosion Operation

Dilation $F \oplus SE$

The value of the output pixel is the maximum value of all pixels in the input pixel's neighbourhood - selected by values of 1 in the structuring element.

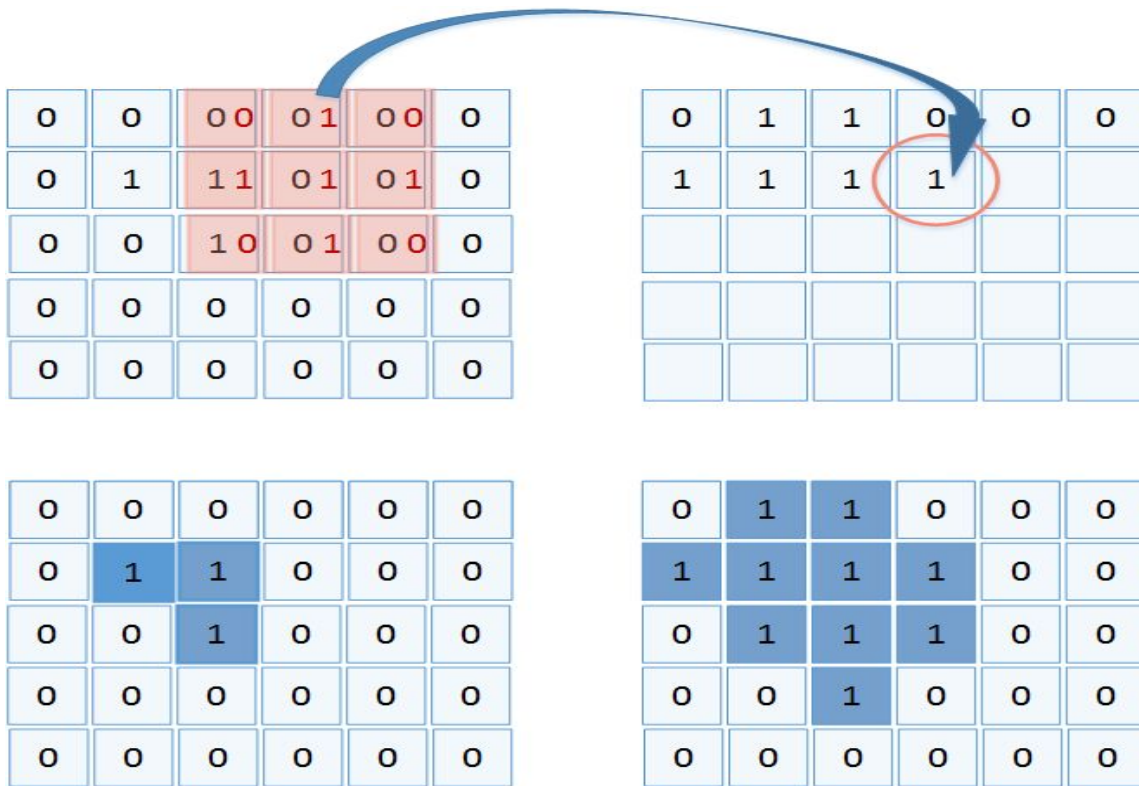
In a binary image, if any of the selected pixels has value 1, the output pixel is set to 1.

Erosion $F \ominus SE$

The value of the output pixel is the minimum value of all pixels in the input pixel's neighbourhood - selected by values of 1 in the structuring element.

In a binary image, if any of the selected pixels has value 0, the output pixel is set to 0.

Example of dilation on a binary image



Binary dilation - example

Input binary image



Image after dilation



Applications of dilation for bridging gaps in an image

Input binary image

Image after dilation

input



dilate



Str. elem.:

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Dilation - an example

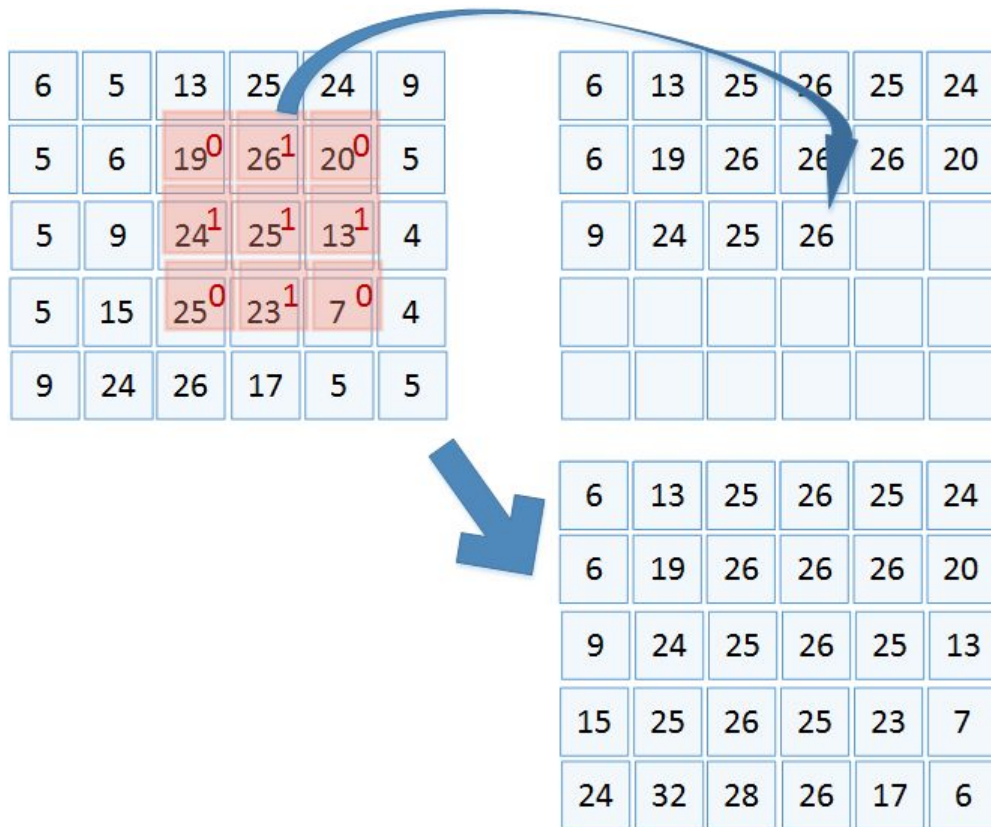
Effect of dilation using the 7×7 structuring element:



structuring
element:

0	0	0	1	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
1	1	1	1	1	1	1
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	1	0	0	0

Example of dilation on a gray-scale image



Example of dilation on a gray-scale image

Input image



Image after dilation



Applications of dilation

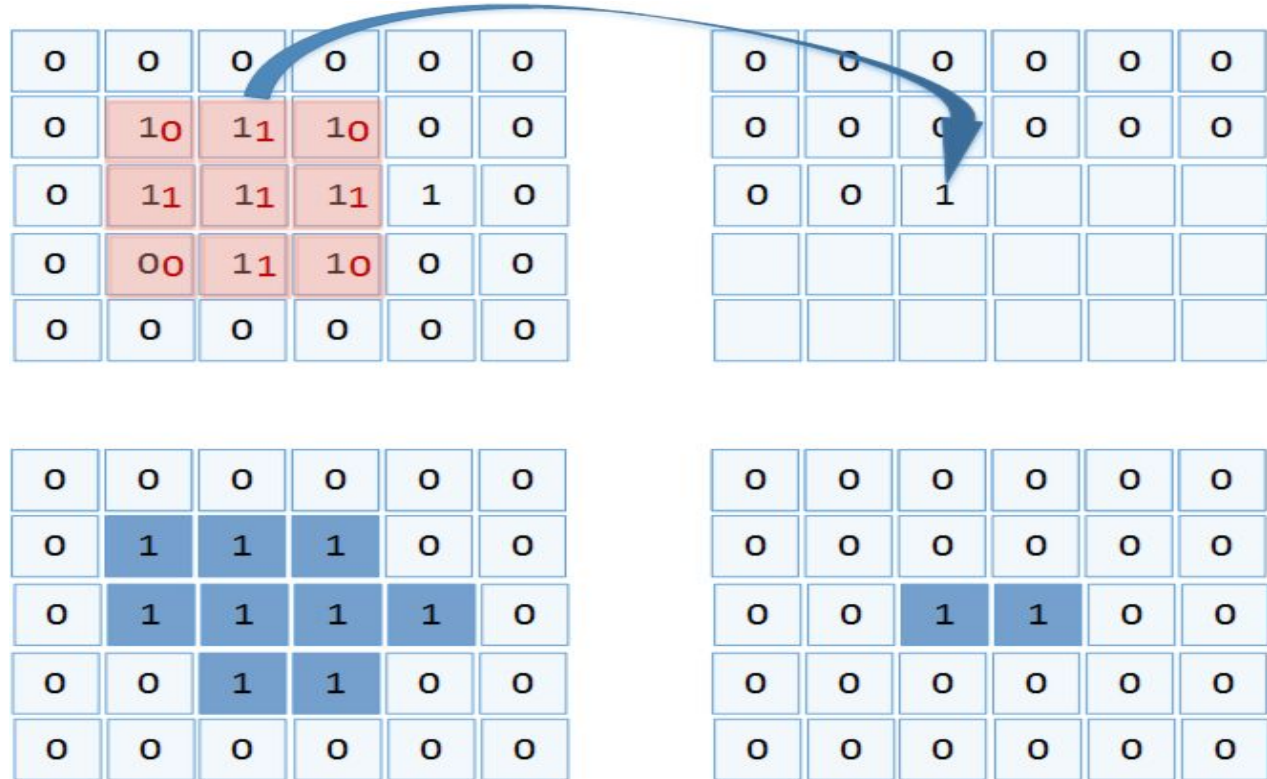
Expand / enlarge white objects in the image

Fill gaps or bays of insufficient width

Fill small holes of sufficiently small size

Connects objects separated by a distance less than the size of the neighbourhood

Example of erosion on a binary image



Applications of erosion

Cut back / decrease white objects in the image

Thus areas of foreground pixels shrink in size,
and holes within those areas become larger.

input



erode



Str. elem.:

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Binary erosion - example

Input binary image



Image after erosion



Examples of Erosion

original image



eroded image with 3 by 3



eroded image with 5 by 5



eroded image with 7 by 7



Erosion - example

Applications of erosion : Eliminating unwanted detail



structuring element:

0	0	0	1	0	0	0
0	1	1	1	1	1	0
0	1	1	1	1	1	0
1	1	1	1	1	1	1
0	1	1	1	1	1	0
0	1	1	1	1	1	0
0	0	0	1	0	0	0

Example of erosion on a gray-scale image

6	5	13	25	24	9
5	6	19 ⁰	26 ¹	20 ⁰	5
5	9	24 ¹	25 ¹	13 ¹	4
5	15	25 ⁰	23 ¹	7 ⁰	4
9	24	26	17	5	5

5	5	5	13	9	5
5	5	6	19	5	4
5	5	9	13		



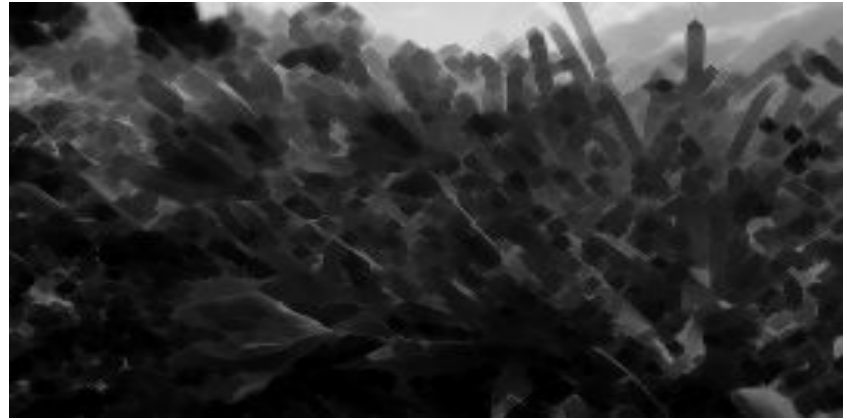
5	5	5	13	9	5
5	5	6	19	5	4
5	5	9	13	4	4
5	5	15	7	4	4
9	18	11	5	5	5

Example of erosion on a gray-scale image

Input image



Image after erosion



Relating Erosion & Dilation

Erosion & dilation are not exact inverse

Dilation cannot

- recreate protuberances eliminated (removed) by erosion
- recreate small objects eliminated (removed) by erosion

Erosion cannot

- recreate holes filled (removed) by dilation
- recreate gaps or bays filled (removed) by dilation

-> we can use sequences of morph. operations

Opening & Closing

The definition of a morphological **opening**: $(F \ominus S) \oplus S$

an **erosion followed by a dilation** using the same structuring element for both operations.

The related operation, morphological **closing** of an image, is the reverse: $(F \oplus S) \ominus S$

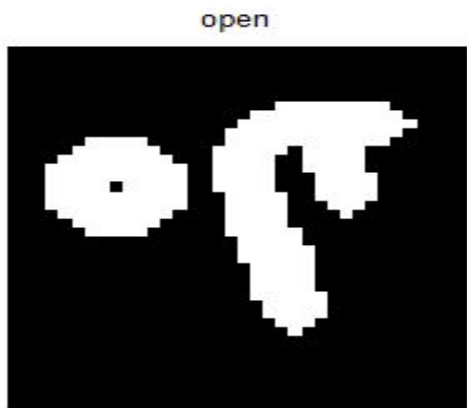
it consists of **dilation followed by an erosion** with the same structuring element.

Example of opening

Input binary image



Image after opening

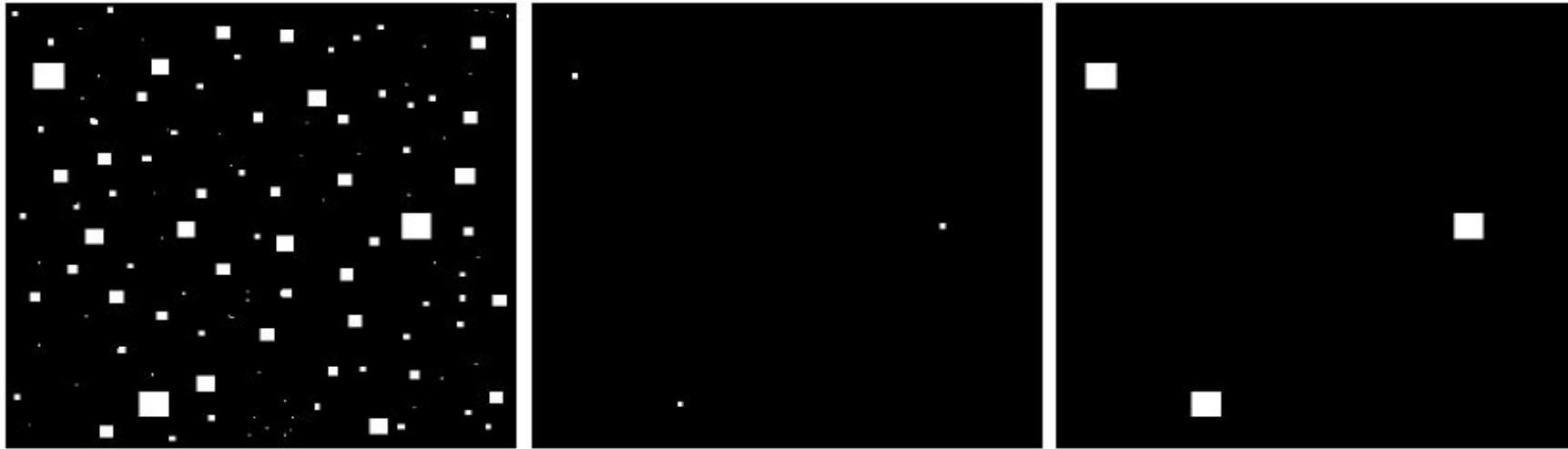


Used structuring element:

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Example of Opening

Remove small objects



a b c

FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.

Example of closing

Input binary image

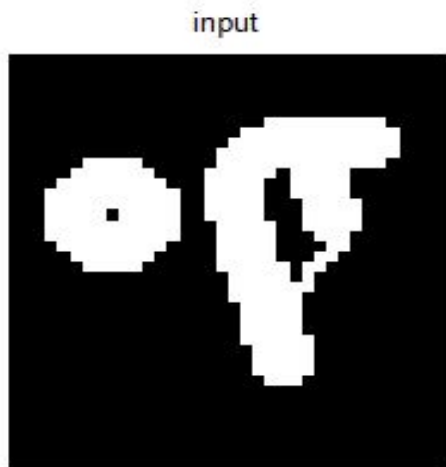
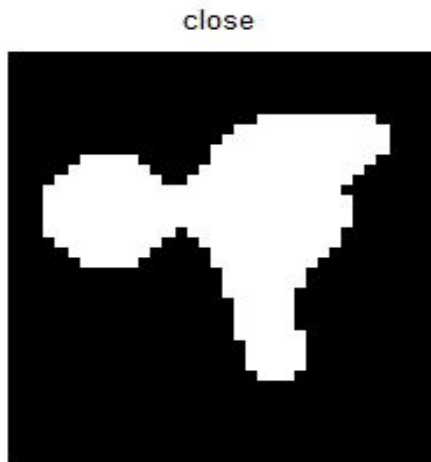


Image after closing



Used structuring element:

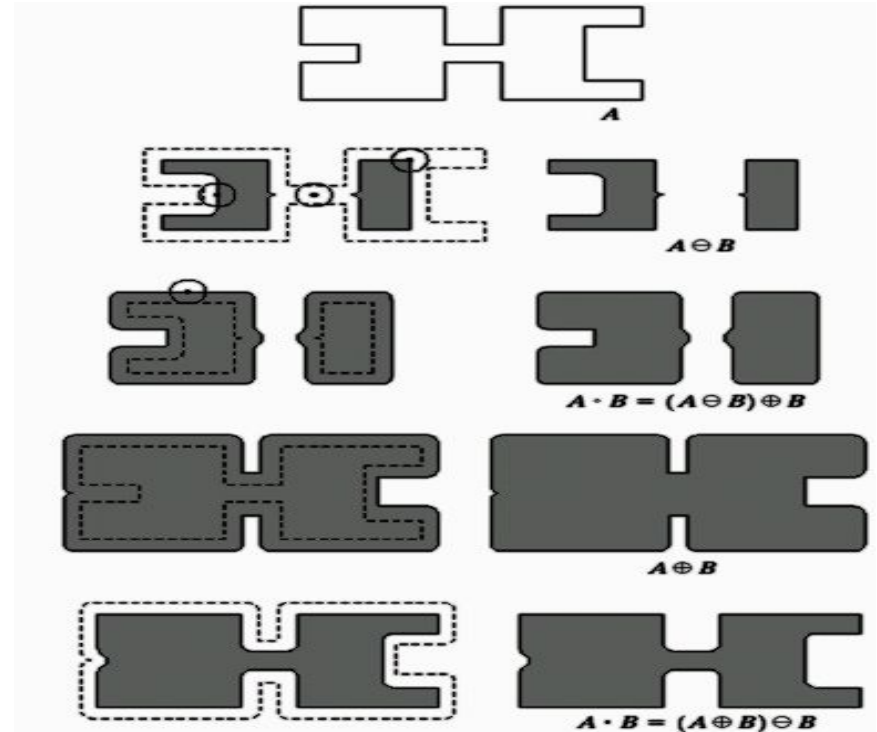
0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Examples

Input image

Example of opening

Example of closing



Source: Digital Image Processing

Rafael C. Gonzalez, Richard E. Woods,

Open & Close



A



opening of A
→ removal of small protrusions, thin
connections, ...



closing of A
→ removal of holes

Examples of binary opening, closing

Input binary image

Open

Close



Distance transform

Every non-zero pixel will be replaced by his distance to the next zero pixel

Input image



Distance transformed image



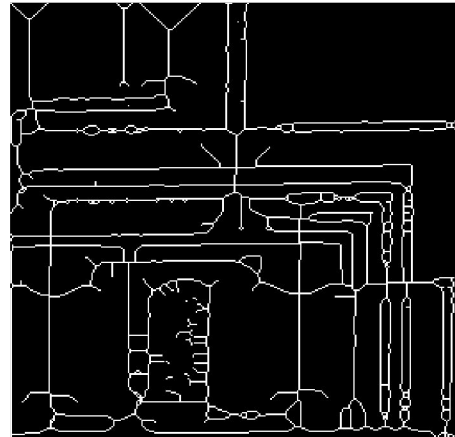
Skeletonization

To reduce all objects in an image to lines, without changing the essential structure of the image

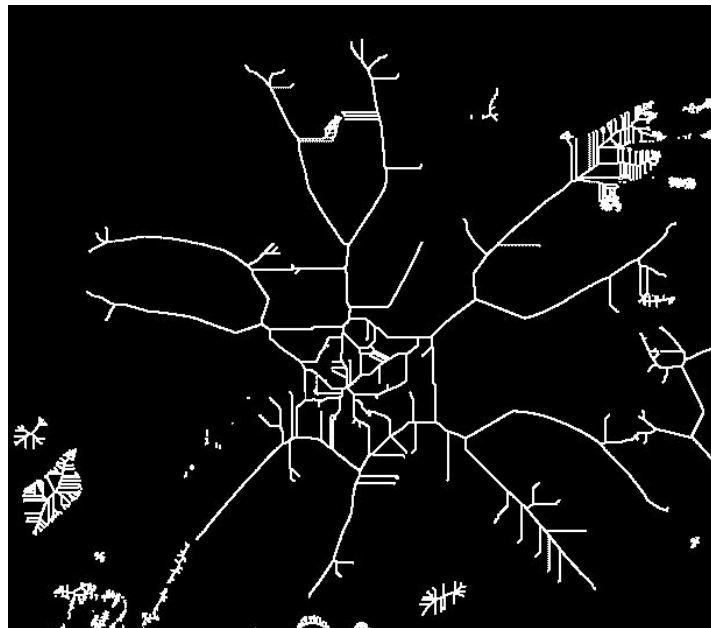
Input Image



Skeleton

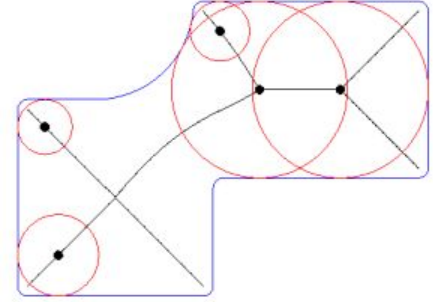


Skeleton



Skeleton algorithms

- algorithm using maximal circles
- sequential morphological thinning
- using distance transform



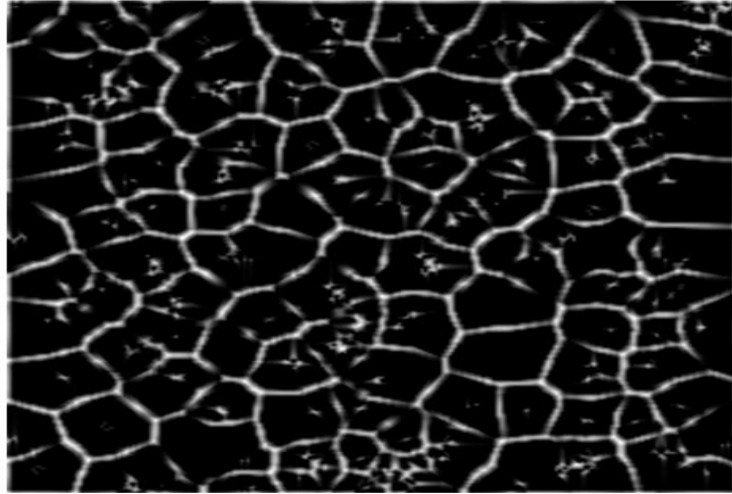
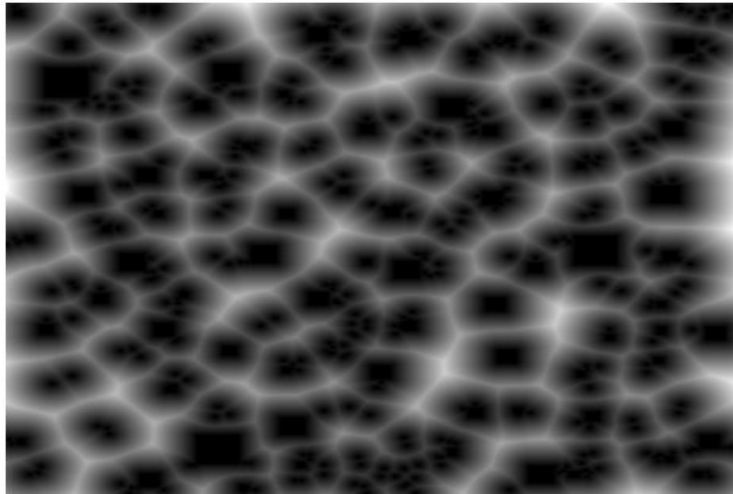
Morphological "top hat" operation

Morphological "top hat" operation returns the image minus the morphological opening of the image.

$$\mathbf{F} - ((\mathbf{F} \ominus \mathbf{S}) \oplus \mathbf{S})$$

Input image

Image after top hat operation



Example of Morphological "top hat" operation

Input Image
operation



Image after top hat



Morphological "bottom hat" operation

$$((\mathbf{F} \oplus \mathbf{S}) \ominus \mathbf{S}) - \mathbf{F}$$

Morphological "bottom hat" operation performs closing (dilation followed by erosion) and then subtracts the original image.

Input image

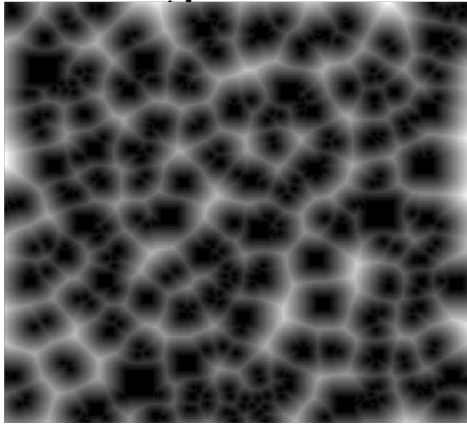
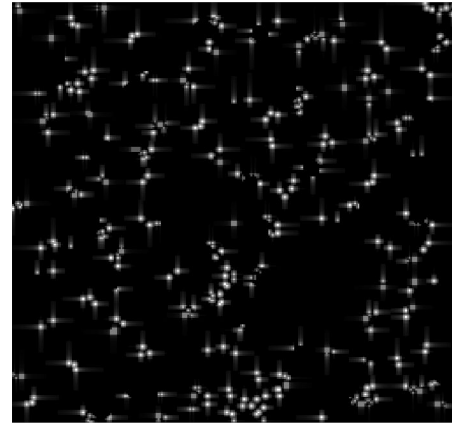


Image after bottom hat

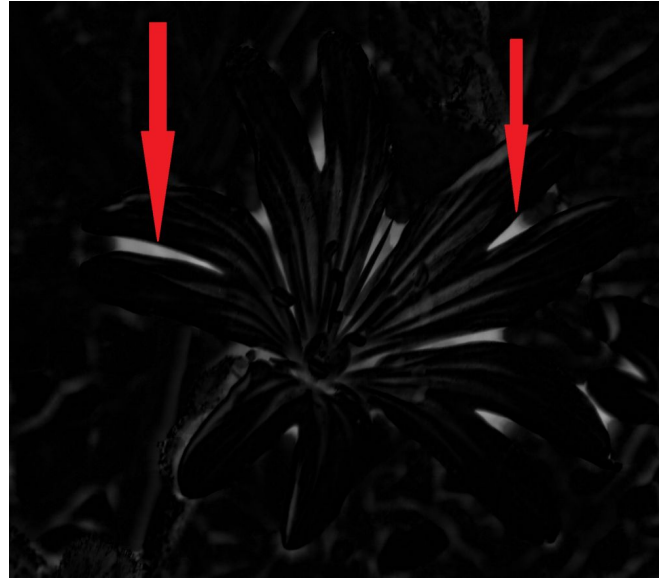


Example of Morphological „bottom hat“ operation

Input Image
operation



Image after bottom hat



Morphological Gradient

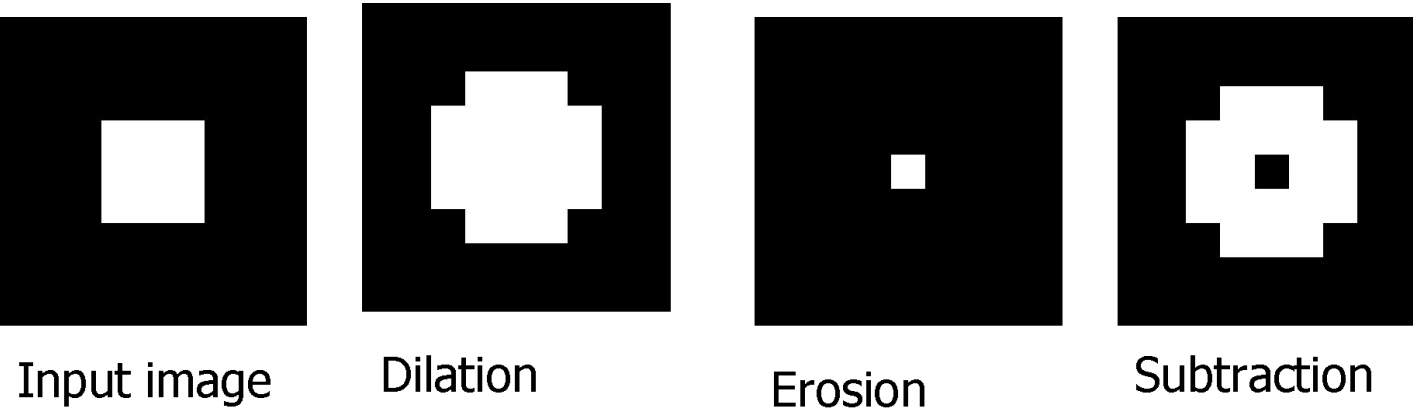
This morphological operator is a composition of a dilation and erosion of the input image by the same structuring element and then the subtraction of these two results.

$$g = (f \oplus b) - (f \ominus b)$$

One important application of the morphological gradient on binary images is the boundary extraction (edge detection).

The same operator can be applied to gray level images.

Morphological Gradient - Example



$$g = (f \oplus b) - (f \ominus b)$$

Str. element:
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Morphological Gradient: An Example of a Gray-scale Image

$$g = (f \oplus b) - (f \ominus b)$$

Input image



Morphological Gradient



Boundary Extraction – Edge detection

Simple boundary extraction:

The boundary is obtained by eroding (or dilating) then taking the difference of the input image and it's erosion (or dilation).

The thickness of the boundary depends on the size of the structuring element.



Hit-or-Miss operation for shape detection

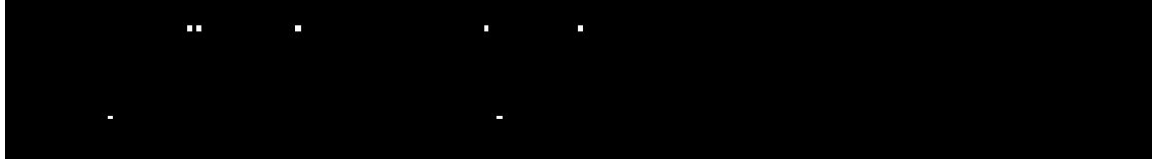
- This operation requires **two structuring elements: SEhm1, SEhm2**. The neighborhoods of those structuring elements SEhm1 and SEhm2 should not have any overlapping elements.
- The hit-or-miss operation preserves pixels whose neighborhoods **match the shape of structuring element SEhm1 and don't match the shape of SEhm2**
- hit-or-miss is defined as the intersection of the erosion of F by the first structure element and the erosion of the complement of F by the second structure element:

$$HM = (F \ominus SEhm1) \cap (F^c \ominus SEhm2);$$

Example of the Hit-or-Miss Operation

We will develop morphological image processing techniques that use combinations and sequences of operations. These techniques are useful in a wide range of applications.

Input image



Result image

Structuring elements:



Example: Hit-Or -Miss

Structuring element:



white pixels(value: +1)

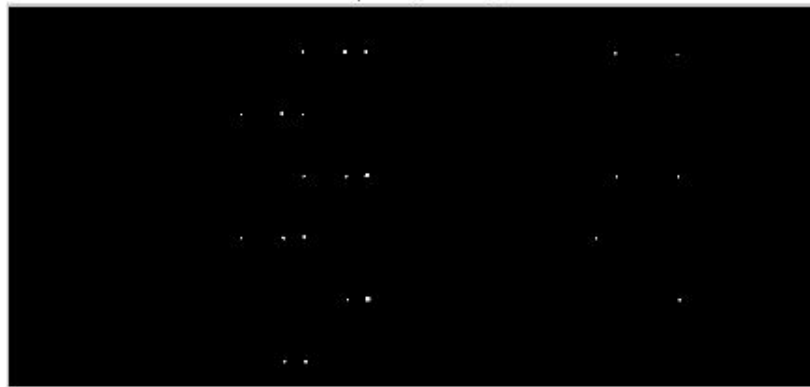
are corresponding to the SEhm1,

black pixels(value : -1)

are corresponding to the SEhm2,

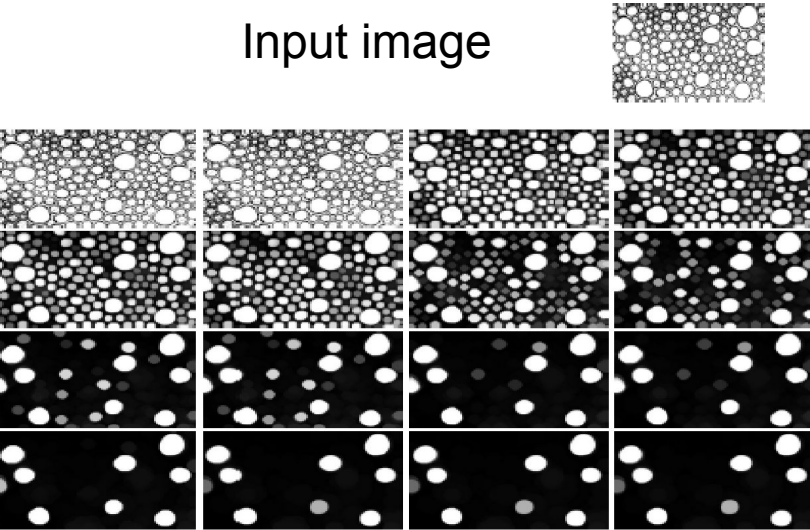
gray pixels (value 0) means ignore.

```
1 CEMENTO DAK63660 dlařba 60x60 I.A  
  DAK636601/IK/8 /  
2 CEMENTO DAK63661 dlařba 60x60 I.A  
  DAK636611/IK/8 /D006  
3 CEMENTO DAKSE660 dlařba 30x60 I.A  
  DAKSE6601/IK/8 /
```

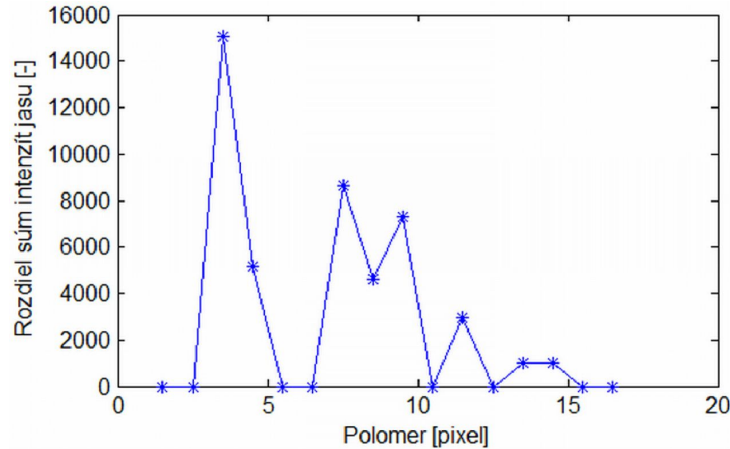


Granulometry example

Input image



Morphological opening with gradually varying size of structural elements.



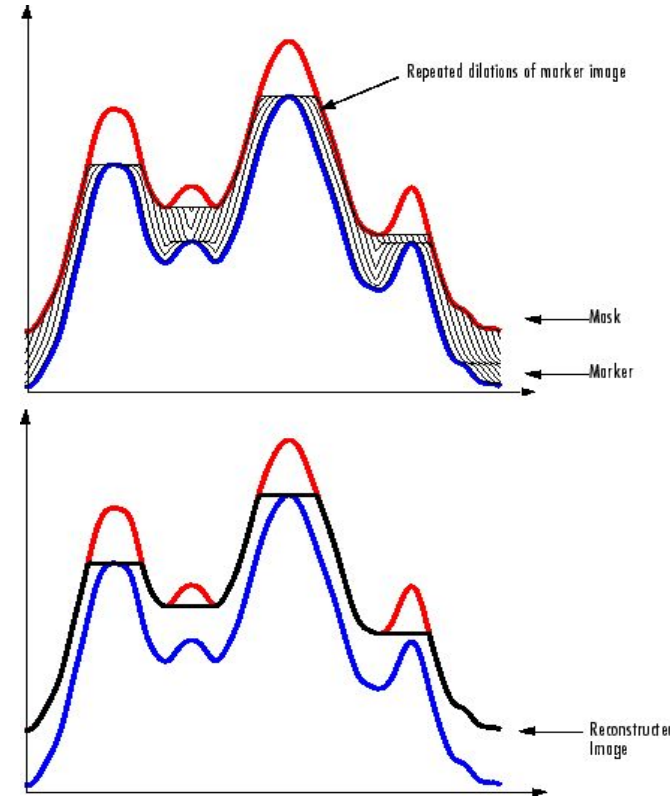
Dependence of the difference of brightness before and after the opening operation by using the morphological structural element of circular shape for varying radius.

Morphological Reconstruction

Morphological Reconstruction

Morphological reconstruction can be thought of conceptually as repeated dilations of an image, called the marker image, until the contour of the marker image fits under a second image, called the mask image.

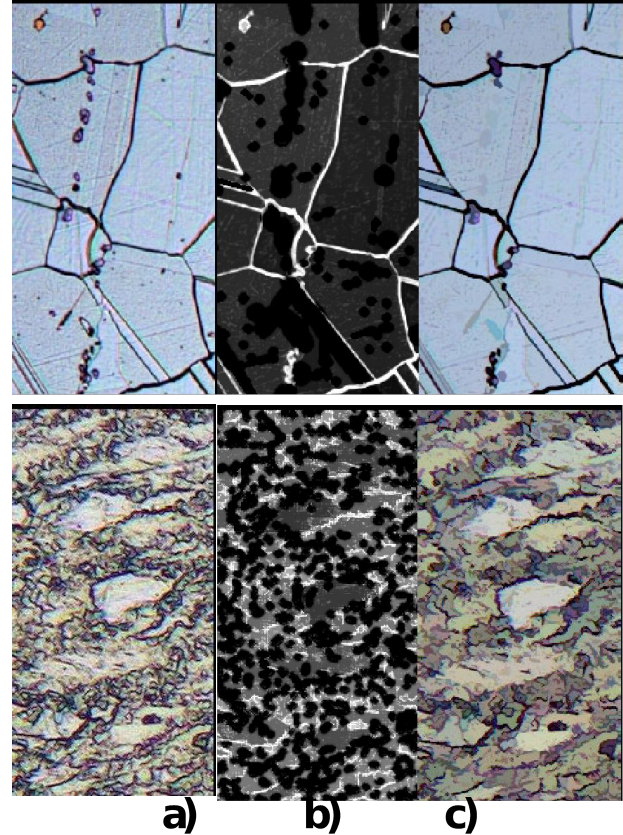
In morphological reconstruction, the peaks in the marker image "spread out,"



Morphological Reconstruction

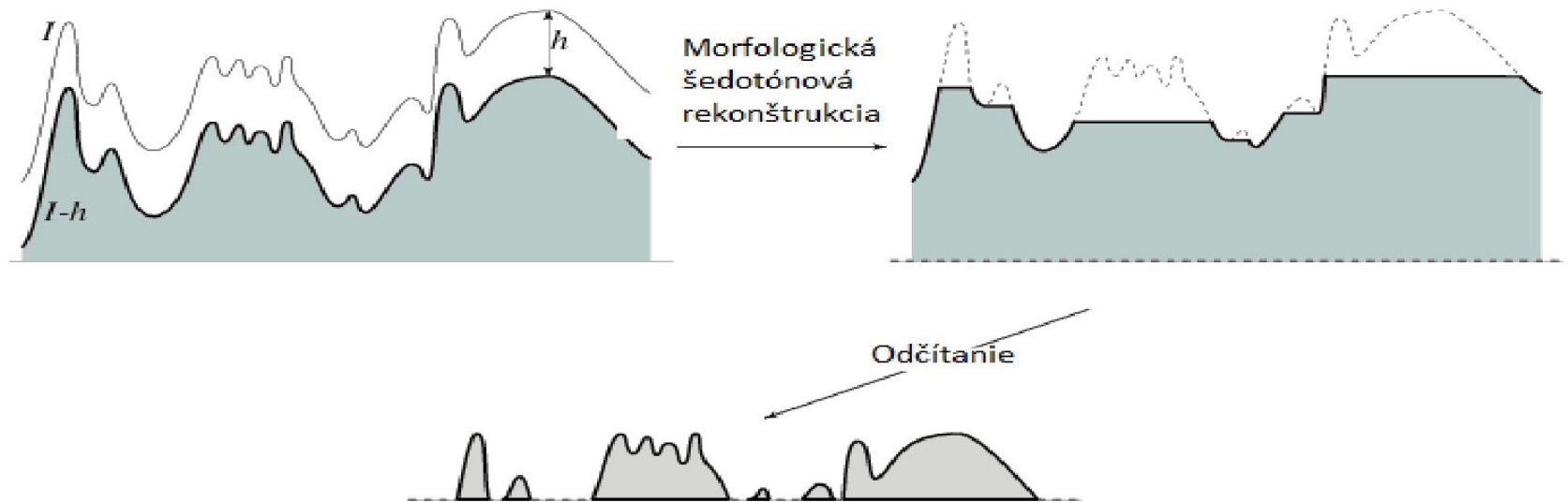
Examples of the image pre-processing and segmentation using morphological reconstruction.

- a) original image,
- b) marker image
- c) reconstructed image



Morphological Reconstruction

Extracting of Local Maxima



Morphological Reconstruction

Extracting of Local Maxima - example

Input image



Local Maxima



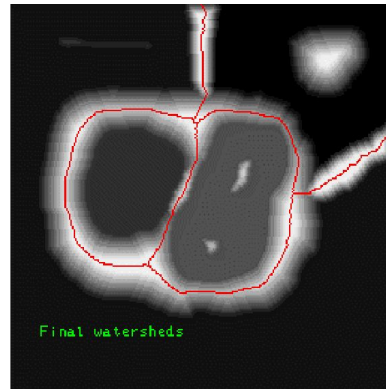
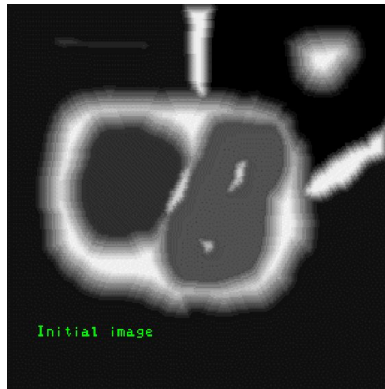
Watershed segmentation

Watershed Transformation

Principle:

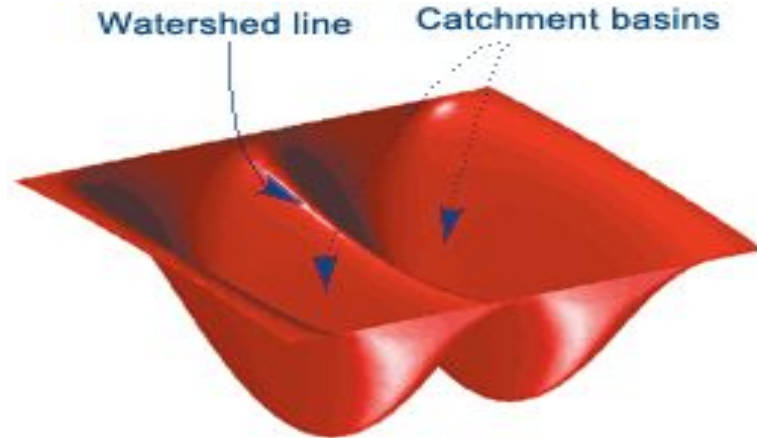
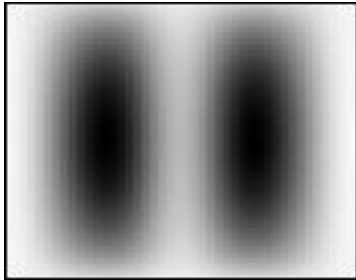
Any grey-scale image can be considered as a topographic surface.

The watershed algorithm is an image processing segmentation algorithm that splits an image into areas, based on the topology of the image.

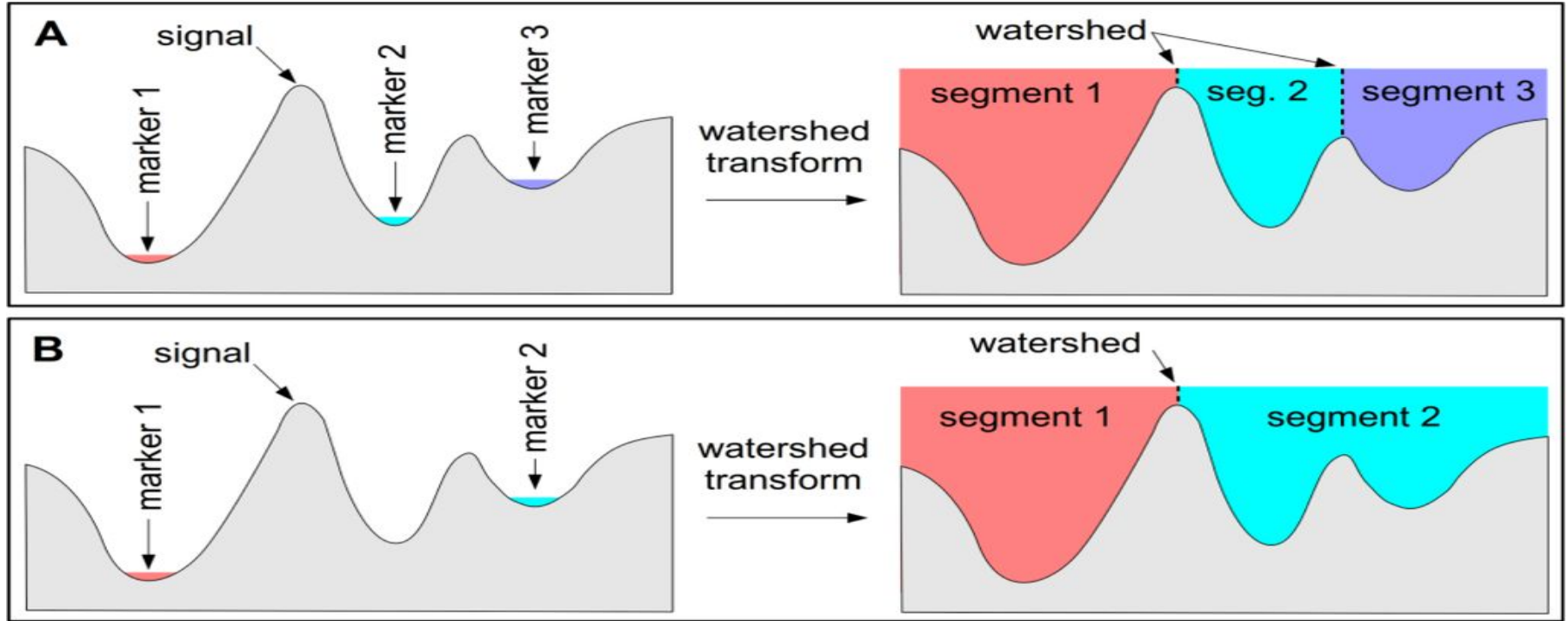


Region based method- Watershed Segmentation

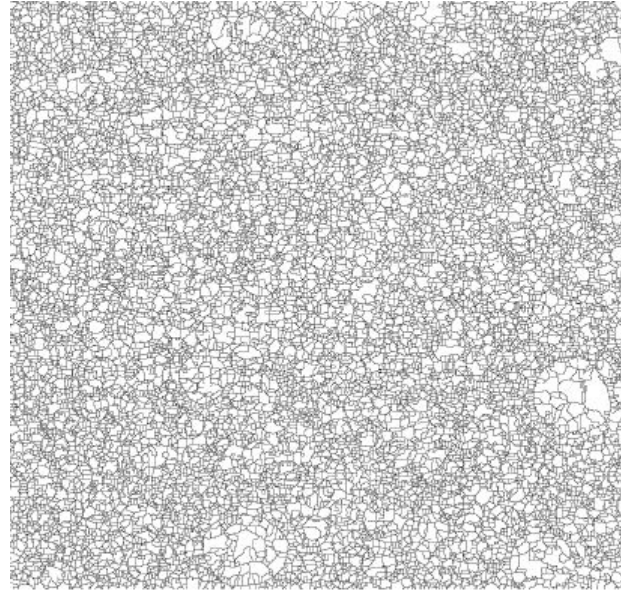
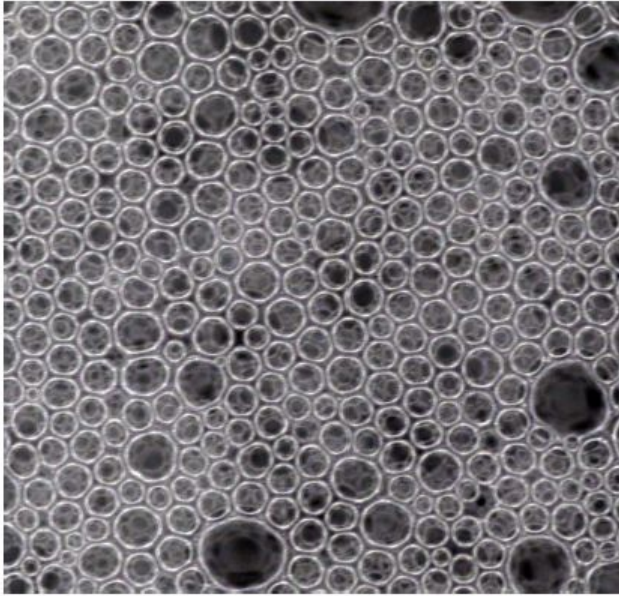
The watershed transform treats the intensity as a function defining 'hills' and 'valleys' and attempts to detect the valleys.



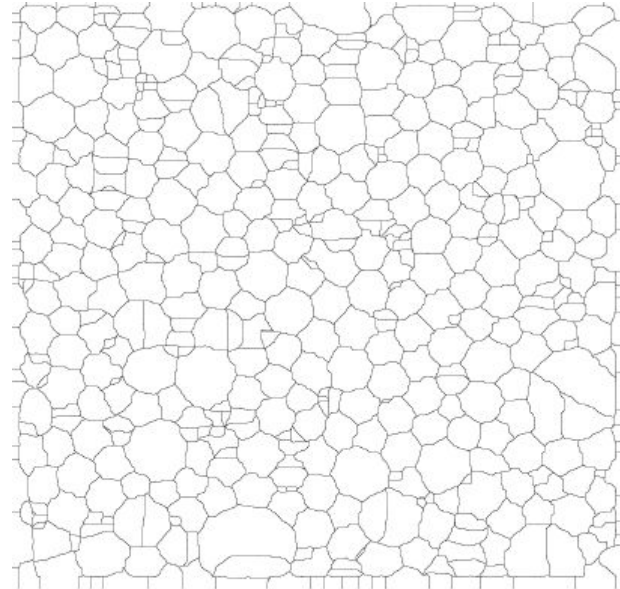
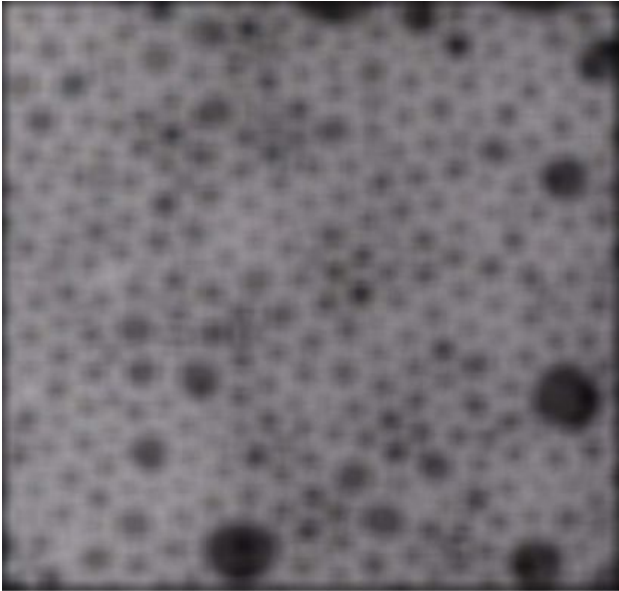
Watershed Transform



Watershed → over-segmentation problem



Watershed after Gauss filtering



Watershed Segmentation for closing the edges

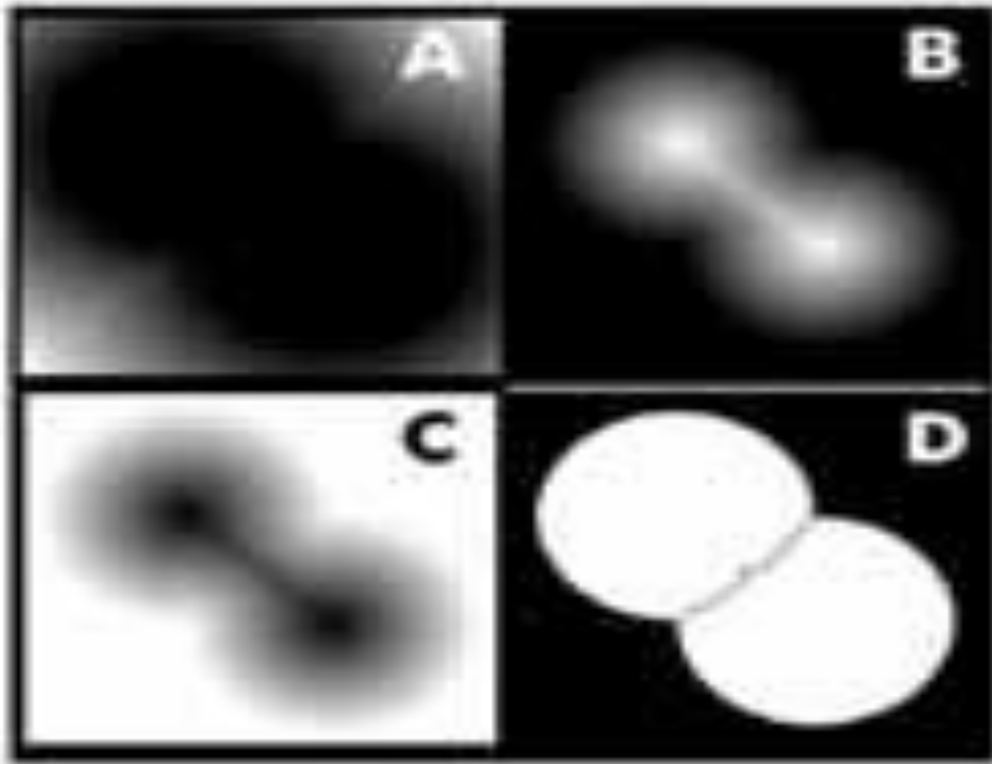
Typical problem: edges are not closed

One possible solution →

Using the methods:

- 1: Distance transform on the edge image
2. Watershed transform on the distance image

Watershed Segmentation for closing the edges



B: Distance transform

C: Images Complement

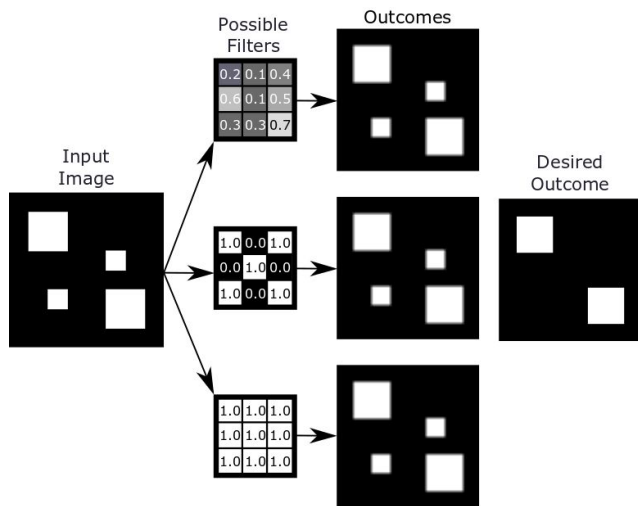
D: Watershed

Deep learning + Morphological processing

Deep learning -> Morphological processing example

Deep Morphological Network (DeepMorphNet)

is able to learn the morphological filters (and consequently the features) based on the input data.

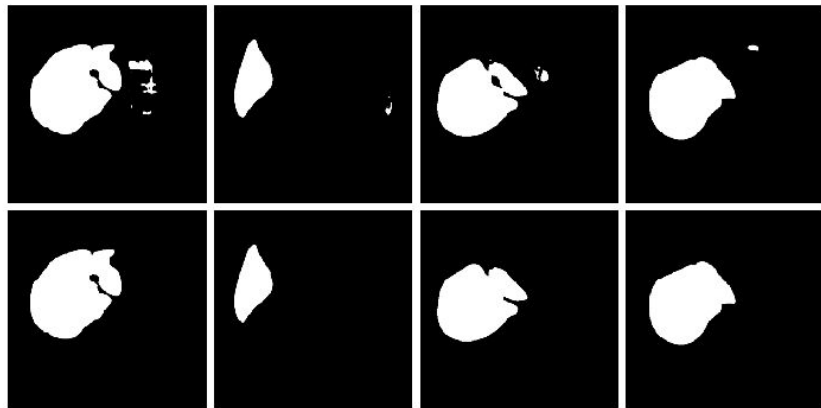


The goal here is to preserve only the larger squares of the input image, as presented in the desired outcome.

The image is processed by three distinct convolutional filters producing different outputs, none of them similar to the desired outcome. However, a simple morphological opening with a 3x3 structuring element is capable of generating such output.

Nogueira K, Chanussot J, Dalla Mura M, Dos Santos JA. An introduction to deep morphological networks. IEEE Access. 2021 Aug 12;9:114308-24.

Deep learning and Morphological Post-processing



The post-processing results using adaptive morphological reconstruction.

Top: segmentation results from the DefED-Net.

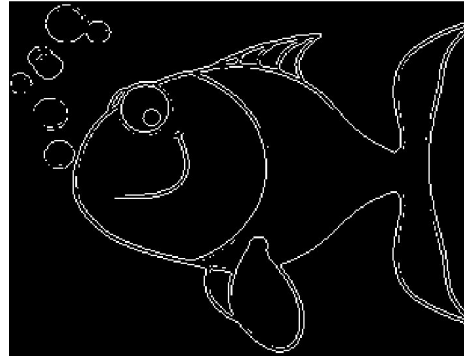
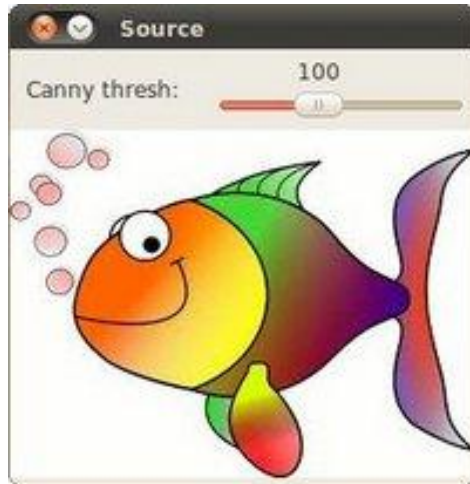
Bottom: post-processing results.

Contours analysis

Contours Finding

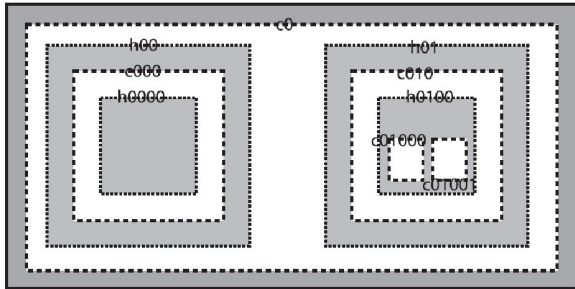
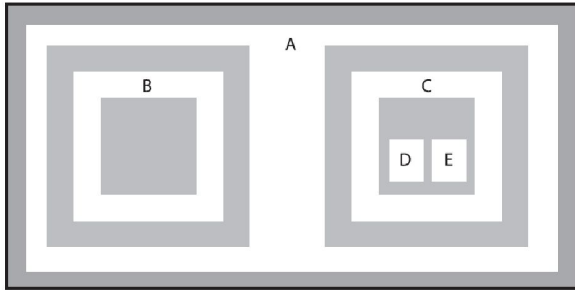
A contour is a list of points that represent a curve in the image.

The contours are a useful tool for shape analysis and object detection and recognition.



Find Contours

OpenCv: findContours()



CV_RETR_EXTERNAL

Retrieves only the extreme outer contours.

CV_RETR_LIST

Retrieves all the contours and puts them in the list.

CV_RETR_CCOMP

Retrieves all the contours and organizes them into a two-level hierarchy

CV_RETR_TREE

Retrieves all the contours and reconstructs the full hierarchy of nested contours.

Find Contours

CV_RETR_EXTERNAL

first = c0

CV_RETR_CCOMP

first = c01000 ↔ c01001 ↔ c010 ↔ c000 ↔ c0
 ↓ ↓ ↓
 h0100 h0000 h01 ↔ h00

CV_RETR_LIST

first = c01000 ↔ c01001 ↔ h0100 ↔ c010 ↔ c000 ↔ h01 ↔ h00 ↔ c0

CV_RETR_TREE

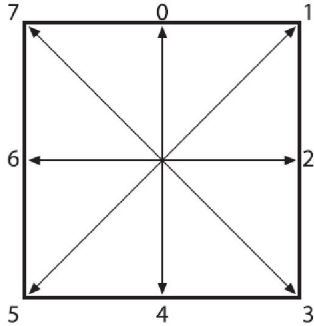
first = c0
 ↓
 h00 ↔ h01
 ↓ ↓
 c000 c010
 ↓ ↓
 h0000 h0100
 ↓
 c01000 ↔ c01001

Find Contours

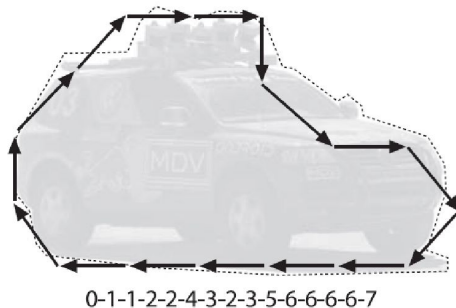
OpenCv: `findContours()`

CV_CHAIN_CODE

Outputs contours in the Freeman chain code



(a)



(b)

a: Freeman chain moves are numbered 0–7;

b: contour converted to a Freeman chain-code representation starting from the back bumper

Find Contours - Structural Analysis and Shape Descriptors

- Approximates a polygonal curve(s) with the specified precision.
- Approximates Freeman chain(s) with a polygonal curve.
- Calculates a contour perimeter or a curve length.
- Calculates the up-right bounding rectangle of a point set.
- Calculates a contour area.
- Tests a contour convexity.
- Finds the convex hull of a point set.
- Finds the convexity defects of a contour.

Find Contours - Structural Analysis and Shape Descriptors

- Fits an ellipse around a set of 2D points.
- Fits a line to a 2D or 3D point set.
- Finds a rotated rectangle of the minimum area enclosing the input 2D point set.
- Finds a circle of the minimum area enclosing a 2D point set.
- Compares two shapes.
- Performs a point-in-contour test.
 - The function determines whether the point is inside a contour, outside, or lies on an edge (or coincides with a vertex).

OpenCv documentation

https://docs.opencv.org/master/d3/dc0/group_imgproc_shape.html#gadf1ad6a0b82947fa1fe3c3d497f260e0